Predicting the Boiling Crisis

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MECH 4V95
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October 30, 2026

I. Problem and Formulation

In nuclear reactors, fuel rods provide an interesting application area for machine learning through the prediction of critical heat flux (CHF). CHF is the heat flux at which the surrounding water can no longer effectively cool the fuel rod. In pressurized water reactors (PWRs), this is commonly caused by departure from nucleate boiling (DNB), where vapor bubbles accumulate near the rod's surface, forming a vapor layer. This vapor layer is less efficient at removing heat from the fuel rod, causing a rapid increase in the rod's temperature. This can lead to "accelerated wall oxidation and potentially to heater (e.g. fuel rod) failure" [1]. Similarly, this can occur in boiling water reactors (BWRs) where the thin liquid film around the fuel rod evaporates, i.e. dryout.

One established method for determining the critical heat flux of a fuel rod is based on a look-up table (LUT) developed on data from over 59 different experiments for vertical, uniformly heated, water-cooled round tubes [2]. However, this LUT has some issues. First, the data it was fitted to has very localized high-density regions and is missing large sections of data. Second, the LUT has very large fluctuations in CHF with pressure, mass flux, quality, and between adjacent entries [3].

As such, machine learning provides a way to develop a novel model for CHF that can describe complex relationships and consider the sparse data regions. Gaussian processes (GP) are suited for this scenario because they return both the predicted value and a predicted uncertainty. This allows the user to assess how confident the model is when given a prediction and can account for that when designing the safety factor of the system. This will be useful in safety analysis for new small modular reactors that have the same safety requirements as traditional large pressurized water reactors and boiling water reactors.

II. Data and Splits

The data used to train the GP models is the same used to develop the LUT. This database contains over 25,000 data points from over 59 individual experiments. From this database, after removing duplicates and unformatted entries, the total number of data points for training is 24,579. Additionally, the heat balance was checked, with the result shown in Figure 1 to validate the data.

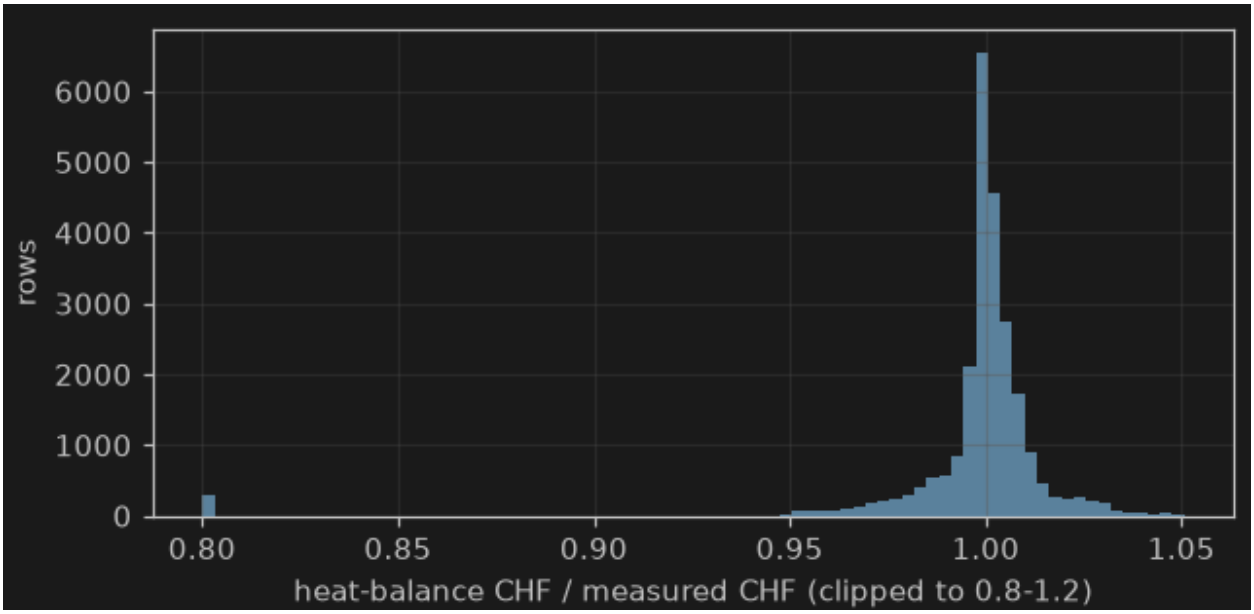


Figure 1: Heat-balance check.

From Figure 1, 98.8% of all rows fall within ± 5% of the predicted heat balance

Additionally, the data's spread across a few of the main inputs can be seen in Figure 2.

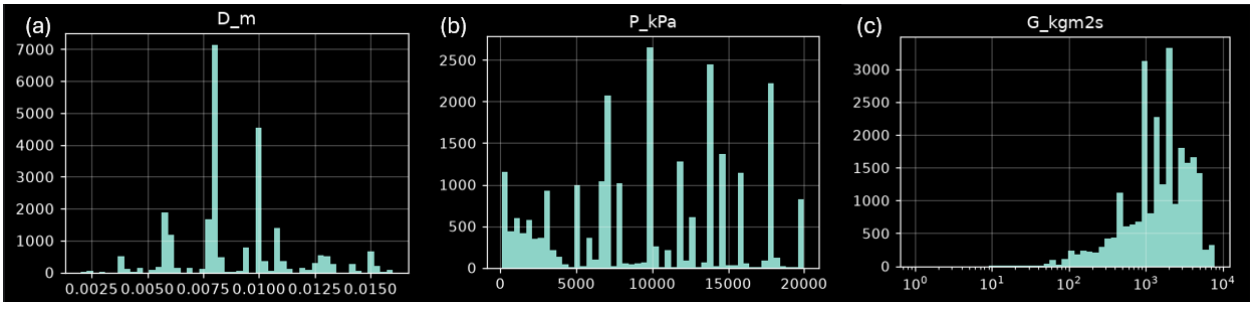


Figure 2: Data distribution graphs. (a) Diameter histogram. (b) Pressure histogram. (c) Mass flux histogram.

Figure 2(a) shows that the dataset does not include any datapoints with a diameter greater than 16mm. Figure 2(b) shows the tendency for the pressure data to come in groups, which is likely due to each of the experimentation results being grouped together around similar pressures. Finally, Figure 2(c) shows the lack of data in low mass flux conditions.

III. Methods

For the GP model, first, an exact GP approach will be evaluated and then a scalable GP (SGPR) approach. These models will use the following input features found in Table 1. The target for the models will be the CHF in log-space. Log-space is used because the CHF covers multiple orders of magnitude and only relative errors are important for training, not absolute error.

Table 1: Input features. Reproduced from [4]

Symbol	Meaning	Units
D	Tube inner diameter	m
P	Pressure	kPa
G	Mass flux	$\frac{kg}{m^2 s}$
x	Thermodynamic equilibrium quality at the CHF location	-

These models will be trained on different splits of the database as seen in Figure 3.

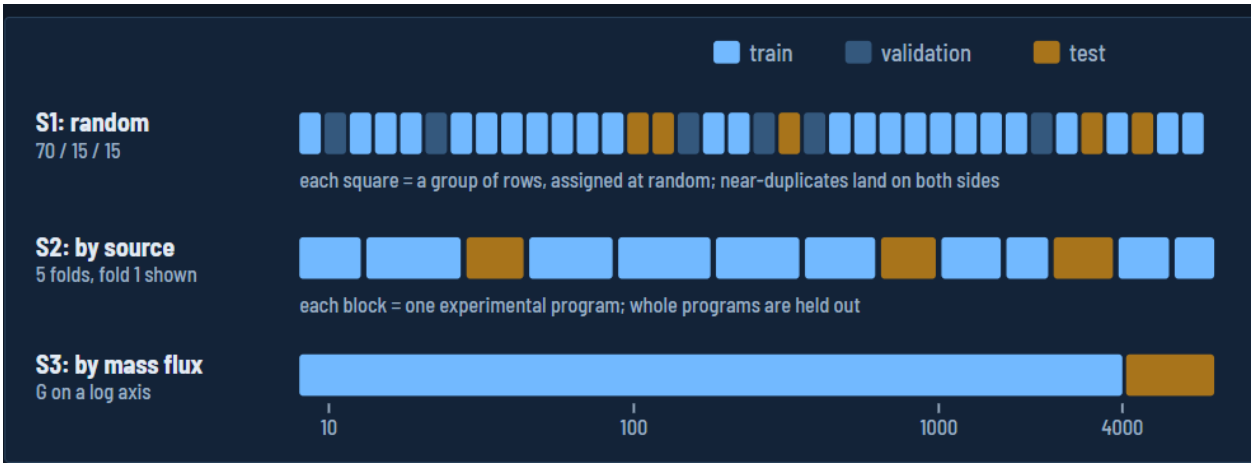


Figure 3: Data splits. [4]

In Figure 3 there are 3 splits for training and evaluation. First, there's split S1 that is from a random selection where 70% is training data, 15% is validation, and 15% is testing. This split is meant for general training and to optimize hyperparameters. The next split is S2 where the dataset is split into 5 folds. Each of these folds contains training data from some experiments and testing data from others. This is meant to show how the model performs when predicting experiments that it has never seen before. Finally, S3 gives all datapoints with mass fluxes less than 4000 as training data and the rest as test data. This split is meant to show how the model performs when making predictions outside the range of its training data.

For exact GP, it will be fit on random subsets of S1 train with n = 500, 2000 and 5000 points. The model will be designed such that the inputs are standardized and the target is in log space. Additionally, it will use a radial basis function kernel (RBF) with automatic relevance determination (ARD), plus a noise term. This model will be trained on the S1 train and hyperparameters will be tuned based on the results of S1 val.

For the scalable GP model, it will use an RBF kernel with ARD. However, this RBF kernel will be inside of an inducing point kernel. This is what makes the scalable GP model scalable. The inducing points create representative datapoints which are then based on the whole dataset. It will be trained and tuned on the same datasets as the exact GP model.

Additionally, each of the models will be trained on the S3 splits to determine how they perform when making predictions outside of the scope of their training data.

As a baseline for model training, an ordinary least squared (OLS) fit was found for each of the splits with results as seen in TABLE II.

TABLE II: OLS performance.

Split	Mean_PM	Std_PM	RMSP_%	MAPE_%	EQ2
S1 val	1.293	1.18	121.5	65.67	0.7683
S1 test	1.255	1.092	112.2	62.33	0.7258
Low-G test	1.567	1.569	166.8	87.27	0.8813
S2	1.283	1.073	112.7	66.68	0.8416
S3 test	1.872	1.021	134.3	98.12	1.67

To show the physical behavior of the models, 4 slices were defined. These slices contain data that only varies one of each of the inputs. These inputs vary over 15 equally spaced values with the ranges shown in TABLE III.

TABLE III: Slice ranges. Reproduced from [4]

Slice	Varies	D(mm)	L(m)	P(kPa)	G $\left(\frac{kg}{m^2 s}\right)$	x
1	D: 2 → 16 mm	–	6.000	14701	998.5	0.391
5	P: 100 → 20,000 kPa	8.00	0.998	–	2006.0	0.140
7	G: 50 → 8000	8.00	1.570	12750	–	0.144
9	x: −0.5 → 1.0	8.14	1.943	9831	1519.5	–

The tuning methods for each of the models are as follows. For the exact GP model, the only hyperparameters changed between runs were the noise level (alpha) and the number of optimizer restarts. First, the alpha level would be varied at a constant number of restarts until the best model based on the validation datasets is determined. Then the number of restarts is increased until the model no longer improves. For the SGPR, the only hyperparameter changed between runs is the number of inducing points. The number of training iterations was left at 100 as this was found to be a good balance between computational cost and model convergence without evidence of overfitting.

IV. Results

Task B1

TABLE IV: 500 samples, 20 restarts.

Length_P	Length_G	Length_D	Length_X	Alpha (3x)	Training time (s)
1.23	2.47	13.4	0.00211	1.87619	9.3

TABLE V: 500 samples split performance.

Split	Mean_PM	Std_PM	RMSP_%	MAPE_%	EQ2
S1 val	1.153	1.822	182.9	37.26	0.2525
S1 test	1.116	1.348	135.3	33.4	0.246
Low-G test	1.317	2.487	250.6	61.04	0.4539

TABLE VI: 500 samples test metrics.

PICP_95	MRIW	NLL_log
0.741	1.017	1.242

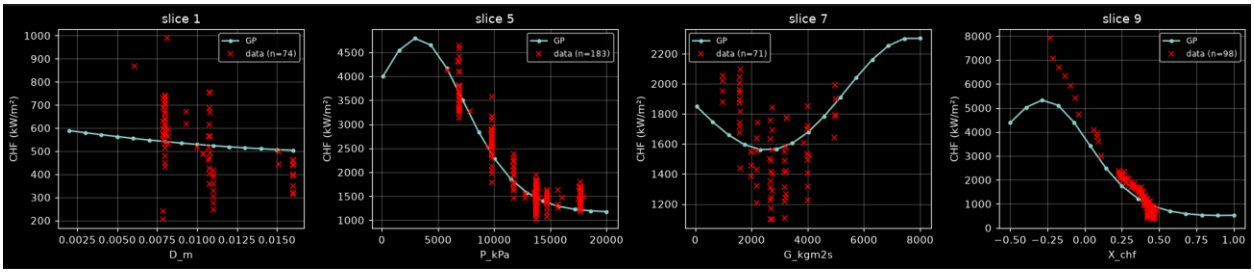


Figure 4: 500 samples slice behavior.

TABLE VII: 2000 samples, 20 restarts.

Length_P	Length_G	Length_D	Length_X	Alpha (6x)	Training time (s)
1.26	2.16	16.8	0.00274	3.91801	143.6

TABLE VIII: 2000 samples split performance.

Split	Mean_PM	Std_PM	RMSP_%	MAPE_%	EQ2
S1 val	1.148	1.581	158.7	34.4	0.1667
S1 test	1.124	1.405	141.1	31.69	0.1515
Low-G test	1.31	2.614	263.1	59.07	0.4161

TABLE IX: 2000 samples test metrics.

PICP_95	MRIW	NLL_log
0.6843	0.8972	1.943

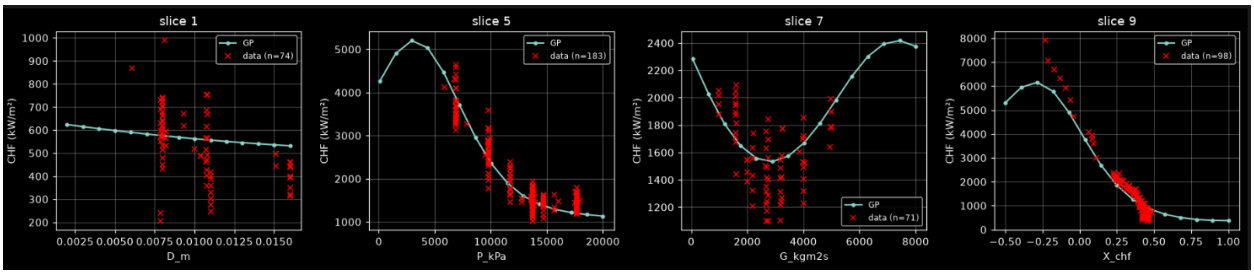


Figure 5: 2000 samples slice behavior.

TABLE X: 5000 samples, 5 restarts.

Length_P	Length_G	Length_D	Length_X	Alpha (0.75x)	Training time (s)
1.14	1.04	3.08	0.0019	0.50253	581.2

TABLE XI: 5000 samples split performance.

Split	Mean_PM	Std_PM	RMSP_%	MAPE_%	EQ2
S1 val	1.079	0.9537	95.68	23.21	0.08457
S1 test	1.062	0.5598	56.31	21.26	0.0698
Low-G test	1.148	0.9759	98.67	35.46	0.2134

TABLE XII: 5000 samples test metrics.

PICP_95	MRIW	NLL_log
0.546	0.3903	6.755

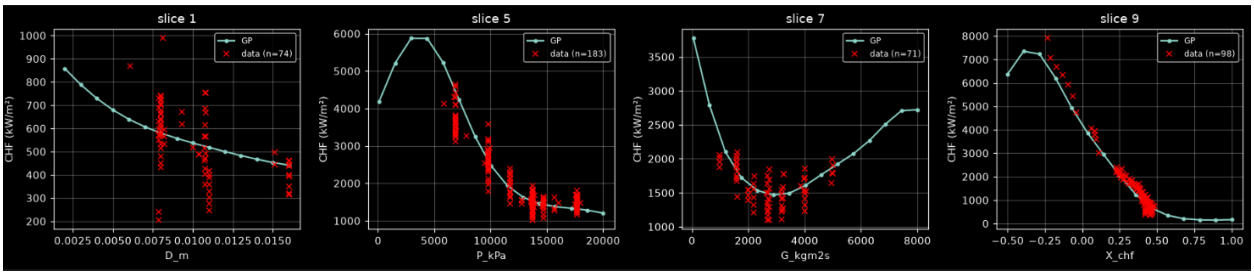


Figure 6: 5000 samples slice behavior.

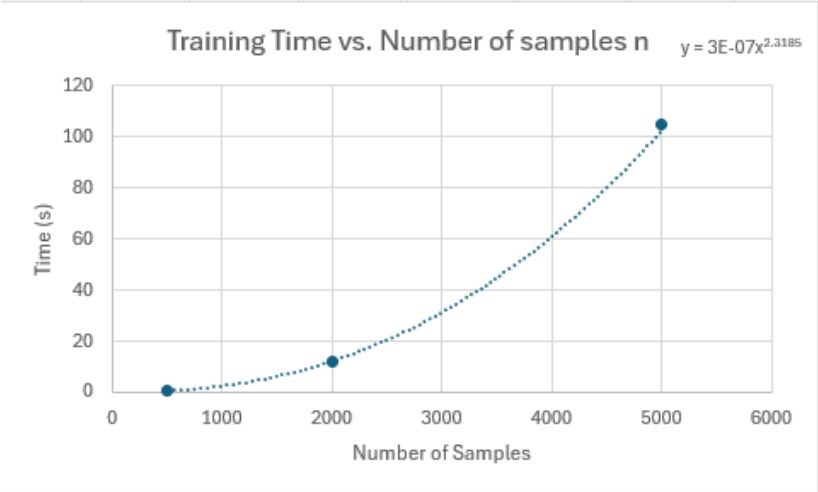


Figure 7: Training time against n.

Task B2 & B3

TABLE XIII: 500 inducing points

Noise	Training time (s)
0.445	25

TABLE XIV: 500 inducing points split performance.

Split	Mean_PM	Std_PM	RMSP_%	MAPE_%	EQ2
S1 val	1.473	2.866	290.4	66.79	0.4234
S1 test	1.401	1.703	174.9	59.47	0.382
Low-G test	1.593	2.983	304	78.57	0.5691

TABLE XV: 500 inducing points test metrics.

PICP_95	MRIW	NLL_log
0.9631	3.27	0.7426

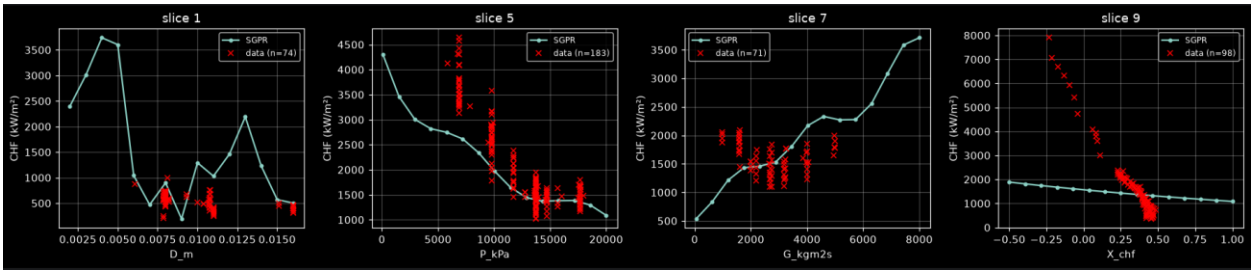


Figure 8: 500 inducing points slice behavior.

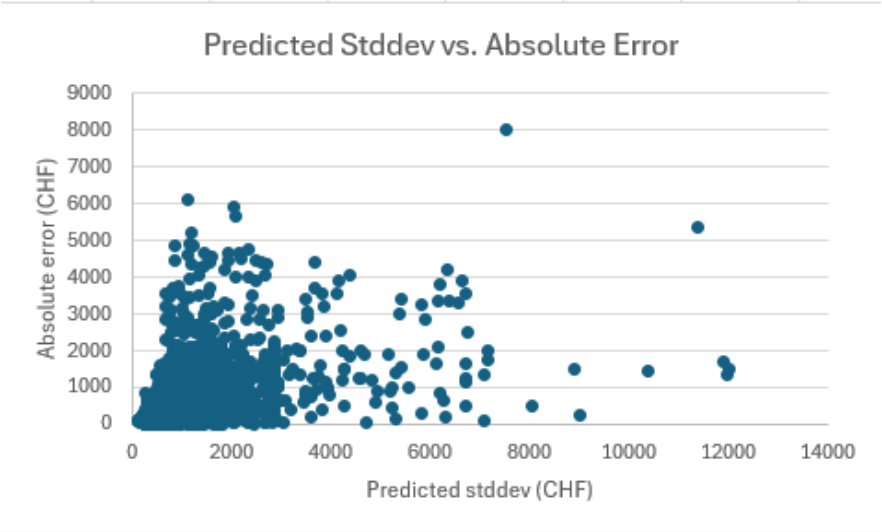


Figure 9: 500 inducing points predicted σ vs. absolute error.

TABLE XVI: 1000 inducing points

Noise	Training time (s)
0.404	88

TABLE XVII: 1000 inducing points split performance.

Split	Mean_PM	Std_PM	RMSP_%	MAPE_%	EQ2
S1 val	1.342	1.162	121.1	53.96	0.3743
S1 test	1.312	0.9064	95.86	50.82	0.3423
Low-G test	1.354	1.259	130.7	55	0.4448

TABLE XVIII: 1000 inducing points test metrics.

PICP_95	MRIW	NLL_log
0.9599	2.782	0.7082

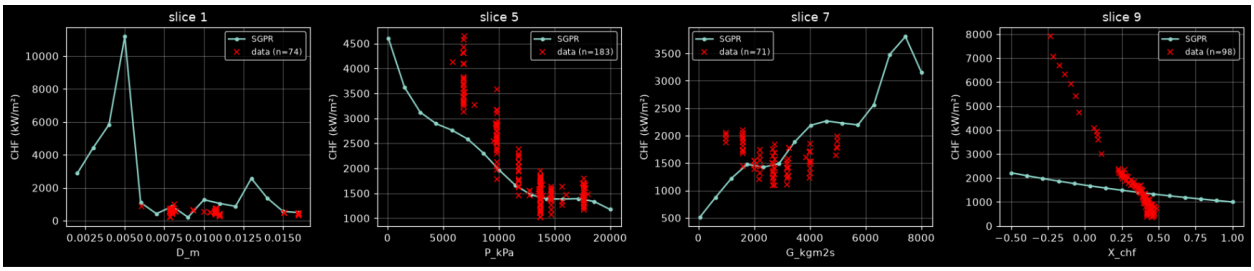


Figure 10: 1000 inducing points slice behavior.

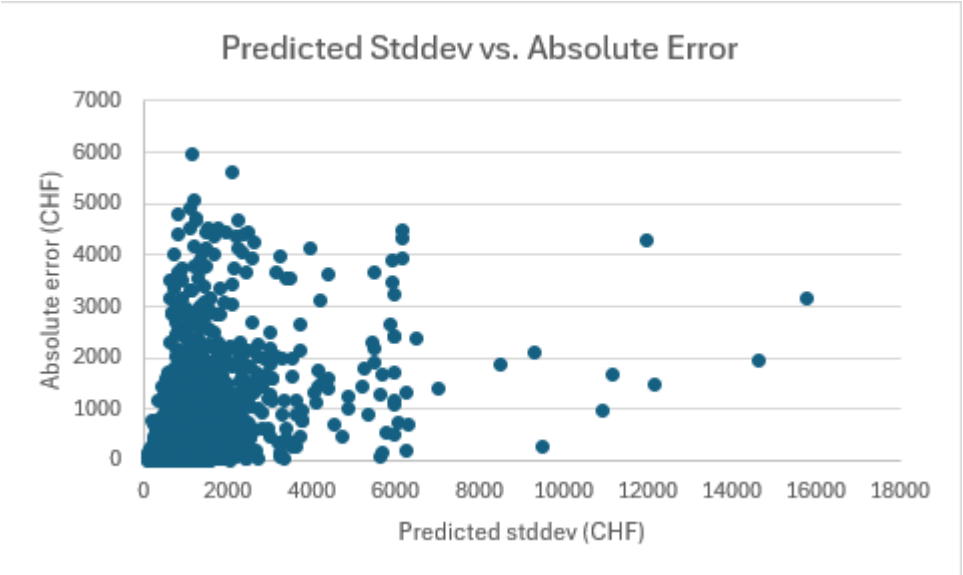


Figure 11: 1000 inducing points predicted σ vs. absolute error.

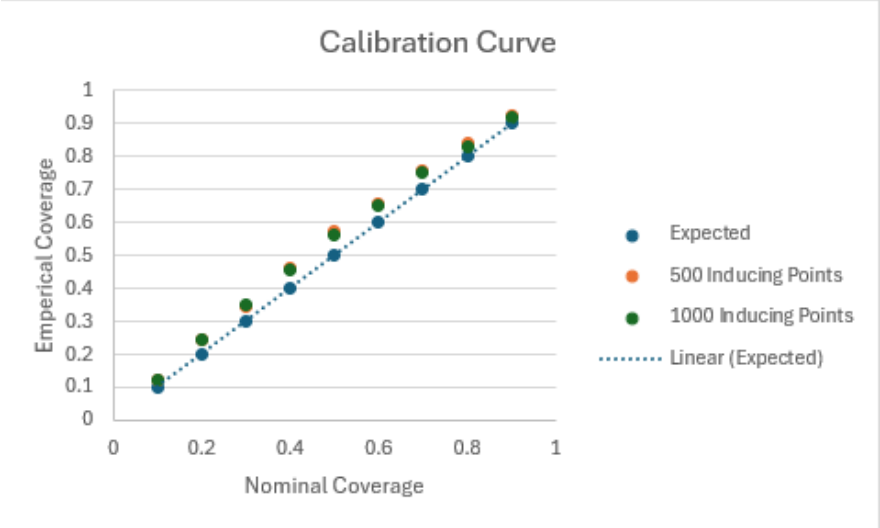


Figure 12: SGPR calibration curve.

Task B4

TABLE XIX: 500 samples, 20 restarts.

Length_P	Length_G	Length_D	Length_X	Alpha (0.75x)	Training time (s)
1.08	2.34	10.9	0.00241	0.45292	16.72

TABLE XX: 500 samples split performance.

Split	Mean_PM	Std_PM	RMSP_%	MAPE_%	EQ2
S3 val	1.066	0.863	86.54	28.74	0.1608
S3 test	0.8473	0.2848	32.31	24.51	0.9044

TABLE XXI: 500 samples test metrics.

PICP_95	MRIW	NLL_log
0.9603	1.383	0.3415

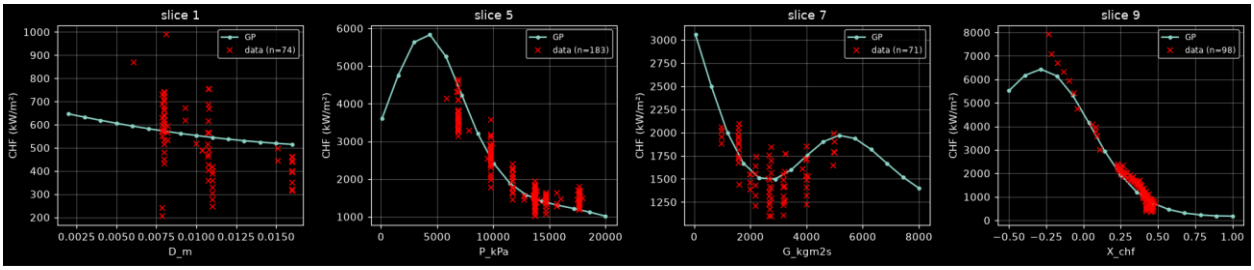


Figure 13: 500 samples slice behavior.

TABLE XXII: 2000 samples, 20 restarts.

Length_P	Length_G	Length_D	Length_X	Alpha (0.75x)	Training time (s)
1.06	1.12	7.48	0.00225	0.45292	229.7

TABLE XXIII: 2000 samples split performance.

Split	Mean_PM	Std_PM	RMSP_%	MAPE_%	EQ2
S3 val	1.072	0.8214	82.45	25.06	0.1115
S3 test	0.7914	0.3258	38.68	30.38	1.127

TABLE XXIV: 2000 samples test metrics.

PICP_95	MRIW	NLL_log
0.901	1.438	0.5787

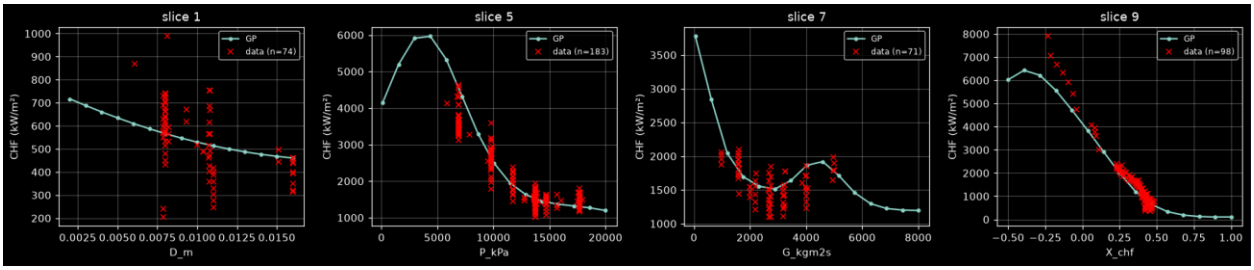


Figure 14: 2000 samples slice behavior.

TABLE XXV: 5000 samples, 1 restart.

Length_P	Length_G	Length_D	Length_X	Alpha (0.75x)	Training time (s)
0.937	1.09	5.09	0.00183	0.45292	322.3

TABLE XXVI: 5000 samples split performance.

Split	Mean_PM	Std_PM	RMSP_%	MAPE_%	EQ2
S3 val	1.057	0.7366	73.87	22.68	0.09219
S3 test	0.8119	0.3285	37.85	29.55	1.075

TABLE XXVII: 5000 samples test metrics.

PICP_95	MRIW	NLL_log
0.8971	1.402	0.5704

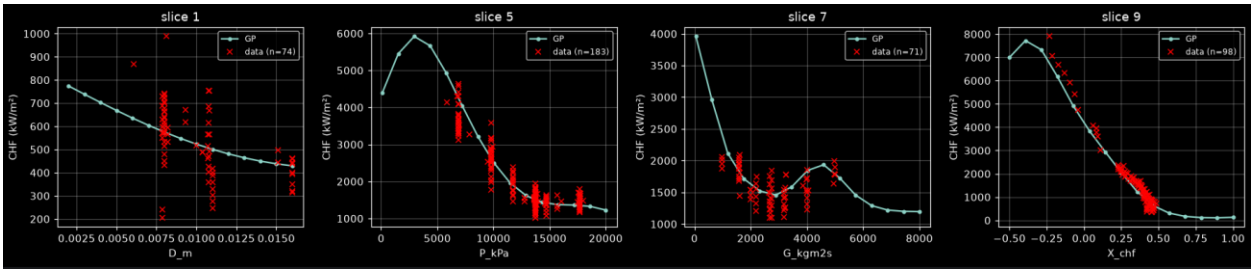


Figure 15: 5000 samples slice behavior.

TABLE XXVIII: 500 inducing points

Noise	Training time (s)
0.439	27

TABLE XXIX: 500 inducing points split performance.

Split	Mean_PM	Std_PM	RMSP_%	MAPE_%	EQ2
S3 val	1.298	0.7355	79.36	51.33	0.4604
S3 test	0.9979	0.5686	56.85	43.45	1.199

TABLE XXX: 500 inducing points test metrics.

PICP_95	MRIW	NLL_log
0.9641	3.744	1.279

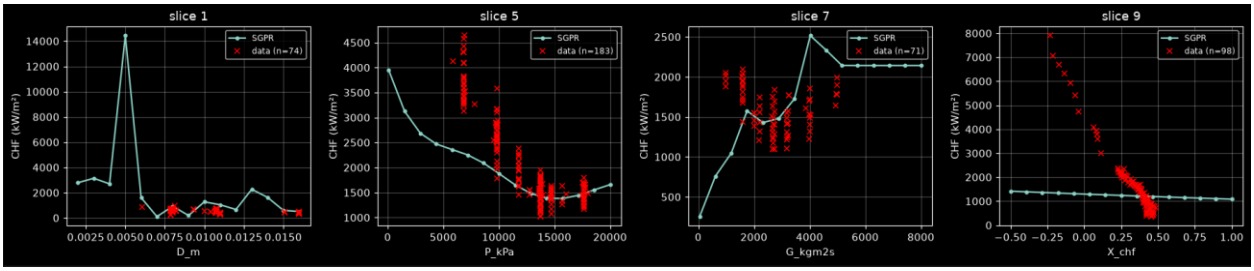


Figure 16: 500 inducing points slice behavior.

TABLE XXXI: 1000 inducing points

Noise	Training time (s)
0.426	94

TABLE XXXII: 1000 inducing points split performance.

Split	Mean_PM	Std_PM	RMSP_%	MAPE_%	EQ2
S3 val	1.289	0.7271	78.25	50.39	0.4488
S3 test	0.9966	0.5663	56.62	43.4	1.198

TABLE XXXIII: 1000 inducing points test metrics.

PICP_95	MRIW	NLL_log
0.9638	3.695	1.269

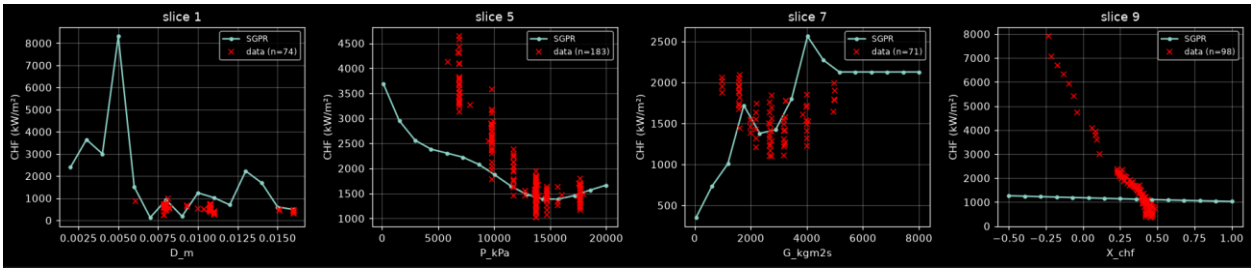


Figure 17: 1000 inducing points slice behavior.

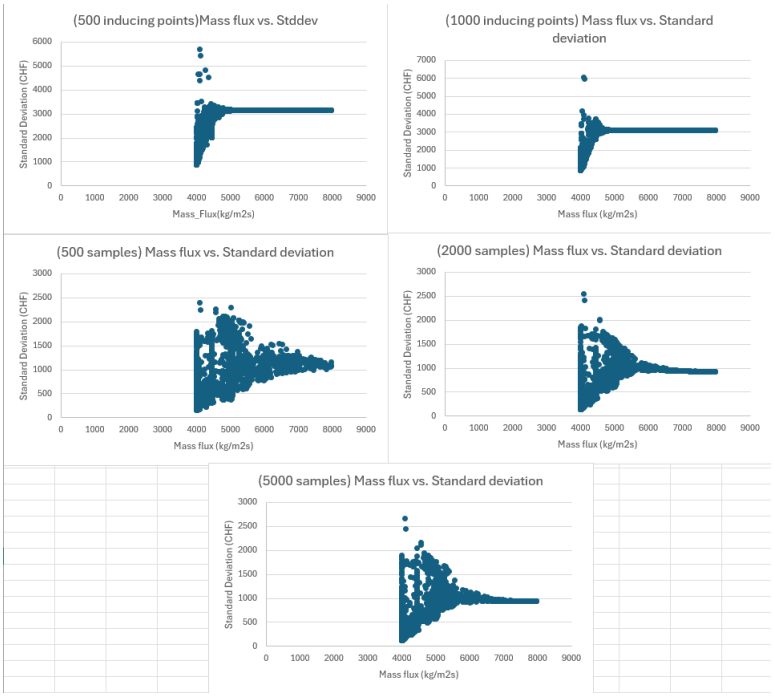


Figure 18: Mass flux vs. Standard Deviation (all models).

V. Discussion

Task B1

From the results of task B1, there were a few noticeable trends. First, the learned length scales typically stayed within the same order of magnitude. This shows that there were no drastic changes in the shape of the input features as the number of training samples increased. Additionally, because ARD was used, the length scales can also reveal how relevant the model thinks each of the features are. The features with shorter length scales like quality have a CHF that changes quickly as the quality changes. On the other hand, features with longer length scales like diameter show smoother changes in CHF as the diameter changes. From this, we can see that the model is more sensitive to quality than the other features when predicting CHF.

As the number of samples the exact GP model was trained on increased, so did the accuracy of the model's predictions. This is evident from how the split performance numbers changed as the number of samples increased. The best model, at 5000 samples had a mean predicted / measured CHF of 1.062. However, there is a very large spread in how well the model predicts CHF and it's overconfident in its predictions. This is evident in the high RMSPE percentages and PICP_95 only covering at most 74% of actual observations. This gets worse as the number of samples increases since the model thinks it's more accurate with more data. Additionally, the slice behavior looks similar across all the sample sizes.

The main hyperparameters changed across each of the runs were the noise level alpha and the number of optimizer restarts. Alpha did not have any relationship with the number of samples and required multiple rounds of iterations before the models behaved. If the alpha was too high or low the model would often increase or decrease 3 our of 4 of the lengths scales towards the kernel limits. This shows that the model is learning to over-rely on one of the features and is overfitting. Finally, it can be seen from Figure 7 that the training time increases at $O(n^{2.3})$, which is close to the expected $O(n^3)$.

Task B2 & B3

Comparing the S1 test results from the split performance tables, exact GP tended to have more accurate mean predictions than SGPR. This is shown from exact GP's lower mean_PM, mean absolute percentage error (MAPE), root mean square percentage error (RMSPE), and EQ2. However, the exact GP struggled with overconfidence. This is shown by how it's means relative internal widths (MRIW) are much lower than SGPR, while the $PICP_{95}$ is only 74% on the best exact model. This suggests that the interval widths are too narrow to capture the variability in the measured CHF. Consequently, despite producing more accurate mean predictions, the exact GP underestimated its uncertainty and failed to cover 95% of the actual observations within its 95% prediction intervals.

The SGPR implementation appears to have much better uncertainty calibration than the exact GP model. First, the PICP_95% for the SGRP is ~95%, as opposed to the exact GP model having a maximum coverage of 74%. Additionally, the negative log-likelihood for SGPR is ~2x lower

than the best exact GP model. This indicates that the SGPR has overall better prediction distribution quality. However, the mean relative internal width for SGPR is higher than in exact GP. This is an expected tradeoff because the width must be higher for the SGPR to appropriately capture 95% in the PICP. Additionally, from the calibration curve, the nominal and empirical coverage remains close across the 10-90% levels. This further supports that the SGPR provides accurate uncertainty estimates.

Task B4

The exact GP model generally performed better than the SGPR when looking at out of distribution performance. The exact GP achieved substantially lower MAPE and RMSPE, however, its mean P/M values were below 1, indicating a tendency for the model to underpredict CHF. Interestingly, increasing the number of samples available to the exact GP model did not improve its performance, only increasing computational cost. Additionally, for both models the standard deviation did not increase as the model moved further away from the training dataset. Instead, the standard deviation appears to converge to a single value as the mass flux increases.

Safety

The way the models are now, none of the GP models in this study should be used for safety analysis. This is because all the models for all the cases in this study have RMSP greater than or equal to ~32%. This is significantly higher than the example neural networks in the OECD/NEA specification. Additionally, their std P/M is greater than or equal to .2848 which is more than double the example value [4].

If they were suitable for safety analysis, the model's mean CHF prediction and prediction uncertainty could be used when evaluating if the departure from nucleate boiling ratio (DNBR) is unsafe. This can be done by ensuring the DNBR does not get too low when compared to the sum of the mean prediction and the prediction uncertainty. The sum of these two values is used in order to consider the model's predicted uncertainty and attempt to ensure the DNBR is based on the worst case scenario.

VI. Limitations

In this study there were many limitations that inhibited the performance of the machine learning models. The first, for exact GP, is that it struggled to contain the 95% within its prediction intervals despite increasing alpha. This could be solved in the future by trying different kernels or even different input features. Additionally, each of the hyperparameters varied in the models was done so manually. This only explores a small space of the hyperparameters. Instead, an automated system like Optuna could be used to determine the optimal hyperparameters. Additionally, the training time for each of the models increases as the number of training datapoints increases. This makes training on larger datasets very computationally expensive. The GP models additionally do not consider the physics of the problem, so it cannot tell if any of the predicted CHF values are not physically possible.

References

[1] J.-M. Le Corre, G. Delipei, X. Wu, and X. Zhao, "Benchmark on artificial intelligence and machine learning for scientific computing in nuclear engineering: Phase 1—Critical heat flux exercise specifications," OECD Nuclear Energy Agency, NEA Working Paper NEA/WKP(2023)1, Jan. 17, 2024.

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[3] D. C. Groeneveld, J. Q. Shan, A. Z. Vasić, L. K. H. Leung, A. Durmayaz, J. Yang, S. C. Cheng, and A. Tanase, "The 2006 CHF look-up table," *Nuclear Engineering and Design*, vol. 237, nos. 15–17, pp. 1909–1922, Sep. 2007, doi: 10.1016/j.nucengdes.2007.02.014.

[4] Y. Xu, "Predicting the boiling crisis," Course project assignment, MECH 4V95, University of Texas at Dallas, Oct. 8, 2026.